UNIVERSITY END-SEMESTER EXAMINATION - 2024

Subject: Operating Systems [CS402] Time: 3 Hours Max Marks: 100

Q1. Explain Banker’s Algorithm for deadlock avoidance with state safety check. [10 Marks]

Q2. State the four necessary conditions for deadlock. How can deadlock be prevented? [8 Marks]

Q3. Explain Dining Philosophers problem and provide a solution using semaphores. [10 Marks]

Q4. Calculate average turnaround time for Preemptive SJF and Round Robin (q=2ms). [10 Marks]

Q5. Explain paging hardware with Translation Lookaside Buffer (TLB). [8 Marks]

Q6. Compute page faults for reference string 7, 0, 1, 2, 0, 3, 0, 4 using FIFO and LRU. [10 Marks]

Q7. Find total head movement for SSTF and SCAN disk scheduling for given cylinders. [8 Marks]

Q8. Differentiate between user-level threads and kernel-level threads. [6 Marks]